FinFlow - Product Analytics Engineering Platform

Event-driven fintech analytics platform modeling customer behavior, transactions, retention, and product growth across savings and investment products.

Source Data Dictionary

Data Classification: Synthetic
Environment: Development
Version: v4.0
Last Updated: 18th July 2026

Created by:
Ajibola Komolafe
Analytics Engineer | Data & Product Analyst
Email: omotoyekomolafe@gmail.com
LinkedIn: LinkedIn Profile
Github: Github Profile
Kaggle: Kaggle Profile

1. **Introduction**

This document defines the source data dictionary for FinFlow, a fast-growing FinTech start-up established in 2016 with operations across the United Kingdom and Ireland. The company offers two core digital financial products: DigiSave, a digital savings product, and DigiVest, a digital investment product.

The primary objective of this project is to simulate an end-to-end, event-driven product analytics and analytics engineering platform that models customer behavior, transactions, retention, and product growth across both savings and investment journeys.

The project begins by generating a foundational dataset spanning a three-year historical period, from the day prior to the data generation date back to three years earlier. To ensure continuity and realism, the dataset is continuously enriched with daily operational data that is appended to the historical baseline. This approach ensures that the dataset remains current, dynamic, and reflective of ongoing product activity.

To enhance realism, business rules and operational constraints have been embedded within the data generation process to closely mirror real-world fintech operations. Detailed specifications of these rules are maintained separately in the data-generation-rules.pdf and business-context.pdf documents.

Overall, this dataset enables comprehensive end-to-end analysis across key business domains, including product growth, user engagement, transaction behavior, funnel conversion, and customer retention.

2. **Document Scope**

This document defines the source datasets generated synthetically for the FinFlow platform. These datasets represent the raw operational data produced by the simulated source systems and serve as the foundation for downstream ingestion, transformation, and analytics processes.

Source data is generated using Python libraries including Faker, NumPy, and Pandas, and is exported as partitioned Parquet files for ingestion into the FinFlow data platform.

The scope of this document includes:

- Source table definitions
- Column-level metadata
- Data types
- Primary and foreign key relationships
- Source system business rules
- Data generation assumptions

This document does not describe warehouse models, transformation logic, derived metrics, or reporting datasets. These are documented separately within the Warehouse Data Dictionary and Business Context Documentation.

3. **Source System Inventory**
The FinFlow source layer consists of the following datasets:

S. No.	Dataset	Type	Description	Grain	No. of Rows
1	dim_date	Dimension	Calendar reference dataset used for analytics and reporting	One record per calendar date	~3,650
2	dim_user	Dimension	Customer profile and registration information	One record per registered user	~500k+
3	dim_product	Dimension	Product catalog covering DigiSave and DigiVest offerings	One record per distinct product offering	2
4	dim_plan	Dimension	Product plan definitions and configurations	One record per product plan variant	4
5	dim_event_type	Dimension	Event taxonomy reference data	One record per distinct event type	16
6	dim_wallet	Dimension	Customer wallet accounts used for funding and transactions	One record per user wallet account (one wallet per user)	~500k+
7	dim_transaction_type	Dimension	Platform transaction type and money movement classification	One record per transaction type	4
8	fact_user_event	Fact	Behavioral event stream capturing customer interactions	One record per user generated event occurrence	~45M
9	fact_transaction	Fact	Transactions related to savings and investment products	One record per money movement transaction within the application	~9M
10	fact_investment_position	Snapshot Fact	Active savings and investment holdings	One record per investment position created by a user	~1.3M+
11	fact_wallet_balance	Snapshot Fact	Latest wallet balance per wallet account	One record per user wallet account	~500k+

4. **Data Refresh Characteristics**

1. **Historical Data**
    - Initial generation covers three years of historical activity.

2. **Incremental Data**
    - New records are generated daily.
    - New user events and financial transactions are appended to the existing datasets.
    - Snapshot fact tables are updated to reflect the latest business state.
    - Slowly Changing Dimension (SCD) Type 1 tables are updated in place to overwrite existing attribute values when changes occur.
    - Change Data Capture (CDC) is used to identify new and modified records during incremental processing.

Table Type	**Load Pattern**
fact_user_event, fact_transaction	Append-only (insert new records)
fact_investment_position, fact_wallet_balance	Insert new records and update existing records (current-state snapshots)
dim_user, dim_wallet	SCD Type 1 (overwrite changed attributes)

3. **File Format**
    - Parquet
4. **Partitioning Strategy**
    - The initial historical backfill is generated as a complete dataset. Subsequent daily incremental loads are written to date-partitioned locations in Azure Blob Storage, enabling efficient ingestion and incremental processing.
    - Event datasets are partitioned by event date.
    - Transaction datasets are partitioned by transaction date.

5. **Naming Conventions**

Source datasets follow the following conventions:

- Table names use snake_case.
- Primary keys are suffixed with "_id".
- Foreign keys reference the primary key of the related dataset.
- Date surrogate keys follow YYYYMMDD format.
- Timestamp columns are stored in UTC.

6. Source Table Definitions

6.1. dim_date

Row Count: ~3,650 rows

Table Description: Calendar dimension containing date-related attributes used for time-based analysis and reporting. Supports daily, weekly, monthly, quarterly, and yearly aggregations across all business processes

Referenced by:
- dim_user
- dim_wallet
- fact_user_event
- fact_investment_position
- fact_transaction

Column Name	Data Type	Description	Example	Nullable
date_id	INT (PK)	Surrogate key representing the calendar date in YYYYMMDD format	20250224	No
date	DATE	The full calendar date	2025-02-24	No
year	INT	Calendar year	2025	No
quarter	INT	Calendar quarter (1–4)	1	No
month	INT	Month number (1–12)	2	No
month_name	VARCHAR (50)	Full month name	February	No
day	INT	Day of month (1–31)	24	No
week_of_year	INT	Week number within the year	8	No
day_of_week	INT	Day of week (0–6)	1	No
day_name	VARCHAR(50)	Full weekday name	Monday	No
is_weekend	BOOLEAN	Indicates whether date falls on Saturday or Sunday	TRUE	No
is_month_start	BOOLEAN	Indicates if date is the first day of the month	TRUE	No
is_month_end	BOOLEAN	Indicates if date is the last day of the month	FALSE	No

6.2. dim_event_type

Row Count: ~16 rows

Table Description: Reference dimension containing the catalog of all trackable user events captured within the FinFlow platform. Defines behavioral events across onboarding, engagement, savings, and investment journeys and provides standardized event classifications for product analytics and funnel reporting.

Referenced by:
- fact_user_event

S. No.	Column Name	Data Type	Description	Example	Nullable
1	event_type_id	INT (PK)	Unique identifier for each event type record in the event type dimension table.	2	No
2	event_type_code	VARCHAR (100)	Machine-readable code used to identify the event type in application tracking and analytics systems.	app_login	No
3	event_type_display_name	VARCHAR (150)	Human-readable name of the event type used for reporting and dashboards.	App Login	No
4	event_type_description	VARCHAR(255)	Detailed explanation of what the event represents and when it is triggered in the product.	User logged into the application	No
5	event_type_category	VARCHAR (50)	High-level grouping of events based on business function (e.g., acquisition, engagement, onboarding).	Engagement	No
6	event_type_stage	VARCHAR (50)	Lifecycle stage of the user journey where the event occurs (e.g., acquisition, onboarding, engagement).	Engagement	No
7	is_conversion_milestone	BOOLEAN	Indicates whether the event represents a key conversion milestone in the user journey or funnel.	FALSE	No
8	product_area	VARCHAR(50)	Product area where the event is generated (e.g., savings, investments, shared).	shared	No

6.3. dim_product

Row Count: ~2 rows

Table Description: Product reference dimension containing FinFlow's core product offerings. Used to categorize user activity and investment positions across the DigiSave and DigiVest product lines and support product-level performance analysis.

Referenced by:
- dim_plan

S. No.	Column Name	Data Type	Description	Example	Nullable
1	product_id	INT (PK)	Unique identifier assigned to each product in the product dimension table.	101	No
2	product_name	VARCHAR (50)	System-friendly name or code used to reference the product in backend systems and analytics pipelines.	digisave	No
3	product_display_name	VARCHAR (255)	User-friendly product name displayed in the application, dashboards, and customer-facing interfaces.	FinFlow DigiSave	No

6.4. dim_plan

Row Count: ~4 rows

Table Description: Reference dimension containing the individual savings and investment plans offered under each product. Enables analysis of customer adoption, engagement, and retention across specific savings and investment offerings.

Referenced by:
- fact_investment_position
- fact_user_event

S. No.	Column Name	Data Type	Description	Example	Nullable
1	plan_id	INT (PK)	Unique identifier for each investment or savings plan in the plan dimension table.	102	No

2	product_id	INT (FK referenced dim_product)	Foreign key linking the plan to its parent product in the product dimension table.	101	No
3	plan_name	VARCHAR (50)	System-level code name used to uniquely identify the plan in backend systems.	fixed_save	No
4	plan_display_name	VARCHAR (50)	User-friendly name of the plan shown in the application and customer interfaces.	FinFlow Fixed Save	No
5	plan_category	VARCHAR (50)	Broad classification of the plan based on its financial purpose (e.g., savings, investment, lending).	Savings	No
6	plan_weight	DECIMAL (5,2)	Denotes the investment type distribution per product	0.35	No
7	interest_rate_min	DECIMAL (5,2)	Minimum interest rate offered by the plan under varying conditions or tiers.	8	No

8	interest_rate_max	DECIMAL (5,2)	Maximum possible interest rate achievable under optimal conditions for the plan.	8	No
9	tenure_days	INT	Fixed duration (in days) for which funds are locked or invested in the plan.	120	No
10	is_open_ended	BOOLEAN	Indicates whether the plan has no fixed maturity date (TRUE = flexible duration).	FALSE	No
11	liquidity_type	VARCHAR (50)	Defines how easily funds can be accessed (e.g., locked, semi-liquid, liquid).	locked	No
12	early_withdrawal_allowed	BOOLEAN	Indicates whether users are permitted to withdraw funds before maturity.	FALSE	No
13	penalty_rate_pct	DECIMAL (5,2)	Percentage penalty applied on interest or principal for early withdrawal.	2	No
14	risk_level	VARCHAR (50)	Indicates the risk classification of the plan based on expected return variability.	Medium	No

6.5. dim_user

Row Count: ~500k+ rows

Table Description: Customer dimension containing demographic, geographic, acquisition, and onboarding information for all registered users. Serves as the central analytical entity for customer segmentation, acquisition analysis, funnel analysis, retention measurement, and product adoption reporting.

This dimension is implemented as a Slowly Changing Dimension Type 1 (SCD Type 1). Current user attributes such as KYC status, activation status, and last login timestamp are overwritten as changes occur. Historical user activities are preserved within the event fact tables.

Referenced by:

- fact_investment_position
- fact_user_event
- fact_wallet_balance

SCD Type: Type 1

S. No.	Column Name	Data Type	Description	Example	Nullable
1	user_id	INT (PK)	Unique surrogate identifier assigned to each user record.	2	No
2	first_name	VARCHAR (50)	User's first name captured during account registration.	Pauline	No
3	last_name	VARCHAR (50)	User's last name captured during account registration.	Morgan	No
4	country	VARCHAR (50)	Country where the user resides or registered from.	UK	No
5	region	VARCHAR (50)	Administrative region or state associated with the user's location.	Scotland	No
6	city	VARCHAR (50)	City associated with the user's registered location.	Edinburgh	No
7	email_address	VARCHAR (255)	Unique email address used by the user for account registration and communication.	pauline.morgan18414@example.com	No

8	reported_annual_income	DECIMAL (18,2)	Self-reported annual income captured during onboarding. Used for customer segmentation, financial suitability analysis, and behavioural simulation	65000	Yes
9	acquisition_channel	VARCHAR (20)	Marketing or acquisition source through which the user was acquired. Allowed values include Organic Search, Paid Social, Referral Program, Direct Traffic, Paid Search, and Partnerships.	Paid Social	No
10	customer_persona	VARCHAR (50)	Marketing persona assigned during synthetic data generation. Personas include Starter Investor, Goal-Oriented Saver, Wealth Builder, Active Investor, and Capital Preserver. The persona influences user demographics, financial characteristics, product adoption, and behavioural simulation.	Starter Investor	No
11	kyc_completed	BOOLEAN	Indicates whether the user has successfully completed Know Your Customer (KYC) verification.	FALSE	No
12	date_of_birth	DATE	User's date of birth provided during onboarding.	1997-02-22	No
13	birth_date_id	INT (FK references dim_date)	Surrogate key referencing dim_date, enabling efficient joins and time-based analysis.	19970222	No
14	signup_date	TIMESTAMP	Date when the user successfully created their account.	2016-01-01 19:25:32	No
15	signup_date_id	INT (FK references dim_date)	Surrogate key referencing dim_date, enabling efficient joins and time-based analysis.	20160101	No
16	is_activated_user	BOOLEAN	Indicates whether the user is eligible for wallet activation and downstream event generation. This field is used internally by the synthetic data generation pipeline.	True	No
17	wallet_activation_timeframe	INT	Time (in minutes) between account signup and first wallet activation. Used only during synthetic event generation and populated only when is_activated_user = TRUE.	14400 (10 Days)	Yes
18	customer_behaviour_se	VARCHAR (50)	Behavioral customer segment assigned during synthetic data generation. The segment is	High_Engagement_	No

	gment		derived from the customer's persona and is used to drive simulated engagement patterns, balance levels, transaction activity, product adoption, retention behaviour, and event generation throughout the platform.	High_Balance	
19	is_immediate_login	BOOLEAN	Indicates whether the user logs into the application immediately after signup. Used internally to simulate different onboarding journeys during synthetic event generation.	True	No
20	supposed_activation_date	TIMESTAMP	This field denotes the pre-populated activation date i.e. first wallet funding date of the user for downstream event creation. This field is created to aid incremental data generation.	2016-01-20 19:25:32	Yes
21	kyc_completion_date	TIMESTAMP	This is a pre-populated field which denotes the supposed kyc completion dates of users that are tagged to complete their KYC competition in the future. This field is used for downstream event analytics. This field is created to aid incremental data generation.	2016-01-18 09:12:54	Yes
22	last_login_at	TIMESTAMP	Stores the timestamp of the user's most recent successful application login. This field is updated whenever an app_login event is generated and provides a current-state view of user engagement for operational reporting, product analytics, and incremental event generation. Historical login activity is retained in fact_user_event.	2026-07-01 02:06:18	Yes

6.6. dim_wallet

Row Count: ~500k+ rows

Table Description: Wallet account dimension representing the primary funding account assigned to each user. Stores wallet-level attributes and supports analysis of wallet activity, funding behavior, transaction flows, and customer activation journeys.

Referenced by:
- fact_investment_position
- fact_user_event
- fact_transaction

- fact_wallet_balance

SCD Type: Type 1

S. No.	Column Name	Data Type	Description	Example	Nullable
1	wallet_id	INT (PK)	Unique identifier assigned to each wallet record in the wallet dimension table.	2	No
2	user_id	INT (FK references dim_user)	Foreign key linking each wallet to the corresponding user in the user dimension table.	2	No
3	wallet_currency	VARCHAR(3)	Denotes the wallet currency	GBP	No
4	wallet_created_at	TIMESTAMP	Timestamp indicating when the wallet was created for the user in the system.	2016-01-01 9:10:25	No
5	wallet_activated_at	TIMESTAMP	Timestamp indicating when the wallet was activated by the user i.e. when the user made the first deposit	2016-02-02 14:25:00	Yes
6	wallet_created_date_id	INT (FK references dim_date)	Date surrogate key representing the wallet creation date for joining with the date dimension table.	20160101	No
7	wallet_activated_date_id	INT (FK references dim_date)	Date surrogate key representing the wallet activation date for joining with the date dimension table.	20160202	Yes

6.7. dim_transaction_type

Row Count: 4 rows

Table Description: Transaction type dimension representing the business classification of financial transactions within the platform. Stores transaction-type attributes and supports analysis of transaction activity, money movement patterns, funding behavior, withdrawals, transfers, and investment-related flows.

Referenced by:
- fact_transaction
- fact_user_event

S. No.	Column Name	Data Type	Description	Example	Nullable
1	transaction_type_id	INT (PK)	Surrogate key uniquely identifying each transaction type.	1	No
2	transaction_type_code	VARCHAR (20)	System-friendly code representing the transaction type.	wallet_funding	No
3	transaction_type_name	VARCHAR (20)	Human-readable name of the transaction type.	Wallet Funding	No
4	transaction_type_description	VARCHAR (255)	Brief description of the business event represented by the transaction type.	Funds deposited into a user's wallet.	No
5	transaction_category	VARCHAR (20)	High-level classification used to group transaction types for reporting and analysis.	Funding	No
6	transaction_direction	VARCHAR (20)	Indicates whether the transaction increases or decreases the wallet balance.	Credit	No

6.8. fact_user_event

Row Count: ~45M+ rows

Table Description: Behavioral event fact table capturing all tracked customer interactions across the FinFlow platform. Records onboarding activities, product usage, engagement events, savings actions, and investment activities. Serves as the primary source of truth for product analytics, funnel analysis, customer journey analysis, and retention measurement.

Referenced by: None

S. No.	Column Name	Data Type	Description	Example	Nullable
1	event_id	INT (PK)	Unique identifier assigned to each event generated in the system.	9000123	No
2	user_id	INT (FK references dim_user)	Identifier of the user who triggered the event.	2001	No
3	event_type_id	INT (FK references dim_event_type)	Classification of the event (e.g., signup_completed, wallet_funded, investment_created).	5	No
4	wallet_id	INT (FK references dim_wallet)	Wallet associated with the event when applicable (e.g., funding, withdrawal, transfer events).	10021	Yes
5	plan_id	INT (FK references dim_plan)	Investment plan selected or activated by the user	2	Yes
6	event_time	TIMESTAMP	Exact timestamp when the event occurred in the system.	2025-05-30 14:22:10	No
7	event_date_id	INT (FK references dim_date)	Date surrogate key used for partitioning and time-based analytics.	20250530	No
8	device_type	VARCHAR (20)	Device platform used by the customer when the event was generated.	ios	No
9	is_money_movement_activity	BOOLEAN	Indicates whether the event resulted in a financial transaction or movement of funds within the platform.	True	No
10	transaction_type_id	INT (FK references dim_transaction_type)	References the transaction type associated with the event when the event results in a financial transaction.	1	Yes
11	transaction_id	INT	Degenerate dimension. Identifier of the transaction produced by this event. Populated	9826	Yes

			only when is_money_movement_activity = TRUE. Not a FK to fact_transaction.		
12	investment_id	INT	Degenerate dimension. Identifier of the investment position created by this event. Populated only when the event type is savings_plan_created or investment_plan_created. Not a FK to fact_investment_position.	63998	Yes

6.9. fact_investment_position

Row Count: ~1.3M+ rows

Table Description: Investment position fact table containing savings and investment holdings created by users. Tracks active and matured positions across savings plans, treasury bills, and mutual funds. Supports analysis of product adoption, portfolio growth, investment behavior, and cross-product usage.

Referenced by: None

S. No.	Column Name	Data Type	Description	Example	Nullable
1	investment_id	INT (PK)	Unique identifier for each investment position created by a user in the system.	10021	No
2	user_id	INT (FK references dim_user)	Foreign key linking the investment to the user who owns it.	2001	No
3	wallet_id	INT (FK references dim_wallet)	Wallet used to fund the investment at creation time.	20001	No
4	plan_id	INT (FK references dim_plan)	Foreign key referencing the investment plan selected by the user.	3	No
5	amount_invested	DECIMAL(18,2)	Principal amount invested by the user in the selected plan.	5000	No
6	investment_start_date	DATE	Date when the investment was initiated and funds were locked into the plan.	2025-01-01	No
7	investment_start_date_id	INT (FK references	Date surrogate key representing the investment start date for time-based	20250101	No

		dim_date)	joins and analytics.		
8	investment_maturity_date	DATE	Expected end date when the investment matures based on tenure.	2026-01-01	No
9	investment_maturity_date_id	INT (FK references dim_date)	Date surrogate key representing the investment maturity date for time-based joins and analytics.	20260101	No
10	investment_status	VARCHAR(20)	Current state of the investment lifecycle. Allowed values include active, redeemed, withdrawn_early.	active	No
11	is_withdrawn_early	BOOLEAN	Flag to denote investments with earlier withdrawal dates than their maturity dates	Null	Yes
12	early_withdrawal_date	TIMESTAMP	Withdrawal dates for investments withdrawn before their maturity date	Null	Yes
13	early_withdrawal_date_id	INT (FK references dim_date)	Date surrogate key representing the early investment withdrawal date for time-based joins and analytics.	Null	Yes

6.10. fact_transaction

Row Count: ~9M+ rows

Table Description: Transaction fact table capturing all monetary movements performed by users within the platform, including wallet funding, withdrawals, investment funding, and investment proceeds transfers.

Referenced by: None

S. No.	Column Name	Data Type	Description	Example	Nullable
1	transaction_id	INT (PK)	Unique identifier assigned to each transaction record within the platform.	1002345	No
2	wallet_id	INT (FK references dim_wallet)	Foreign key referencing the wallet impacted by the transaction.	20012	No
3	transaction_type_id	INT (FK references dim_transaction_type)	Foreign key identifying the type of transaction performed.	1	No

4	transaction_amount	DECIMAL(18,2)	Monetary value associated with the transaction stored as a positive amount.	5000	No
5	transaction_status	VARCHAR(20)	Processing outcome of the transaction (success, failed, reversed).	success	No
6	transaction_timestamp	TIMESTAMP	Exact timestamp when the transaction event occurred within the platform.	2025-04-10 14:32:55	No
7	transaction_date_id	INT (FK references dim_date)	Surrogate key representing the transaction date for joining with the date dimension table.	20250410	No

6.11. fact_wallet_balance

Row Count: 500k+

Table Description: Snapshot fact table containing the latest available wallet balance for each user. The table is updated incrementally whenever a successful financial transaction changes a wallet's balance and represents the current financial state of each wallet.

Referenced by: None

S. No.	Column Name	Data Type	Description	Example	Nullable
1	wallet_id	INT (PK)	Unique identifier for the wallet. One wallet represents one user's cash balance.	1002345	No
2	user_id	INT (FK references dim_user)	Foreign key identifying the user account associated to the wallet	1002345	No
3	current_balance	DECIMAL (15,2)	The current available balance in the wallet after processing all successful transactions.	3,500	No
4	updated_at	TIMESTAMP	Timestamp of the most recent business event or transaction that changed the wallet balance.	2025-04-10 14:32:55	No
5	updated_date_id	INT (FK references dim_date)	Foreign key to the date dimension representing the date on which the wallet balance was last updated.	20250410	No
6	last_transaction_id	INT	Degenerate key referencing the most recent transaction that changed the wallet balance. Used for auditing and lineage.	159836	No

7	last_event_id	INT	Degenerate key referencing the user event that triggered the latest wallet balance update. Used for event lineage and auditing.	450963	No

7. **Source Data Relationships**

Table	**Relationship**
dim_user → dim_date	**Many-to-One via signup_date_id & birth_date_id**
dim_plan → dim_product	**Many-to-One via product_id**
dim_wallet → dim_user	**One-to-One via user_id**
dim_wallet → dim_date	**Many-to-One via wallet_created_date_id, wallet_activated_date_id**
fact_investment_position → dim_user	**Many-to-One via user_id**
fact_investment_position → dim_wallet	**Many-to-One via wallet_id**
fact_investment_position → dim_plan	**Many-to-One via plan_id**

fact_investment_position → dim_date	**Many-to-One via investment_start_date_id, investment_maturity_date_id & early_withdrawal_date_id**
fact_transaction → dim_date	**Many-to-One via transaction_date_id**
fact_transaction → dim_wallet	**Many-to-One via wallet_id**
fact_transaction → dim_transaction_type	**Many-to-One via transaction_type_id**
fact_user_event → dim_user	**Many-to-One via user_id**
fact_user_event → dim_event_type	**Many-to-One via event_type_id**
fact_user_event → dim_wallet	**Many-to-One via wallet_id**
fact_user_event → dim_date	**Many-to-One via event_date_id**
fact_user_event → dim_plan	**Many-to-One via plan_id**

fact_user_event → dim_transaction_type	**Many-to-One via transaction_type_id**
fact_user_event.transaction_id	Degenerate dimension. Carries the identifier of the transaction produced by the event. Populated only when **is_money_movement_activity** = **TRUE**. No referential join to **fact_transaction** and used for filtering and cross-referencing only.
fact_user_event.investment_id	Degenerate dimension. Carries the identifier of the investment position created by the event. Populated only when the event type is **savings_plan_created** or **investment_plan_created**. No referential join to **fact_investment_position** and used for filtering and cross-referencing only.
fact_wallet_balance → dim_wallet	**One-to-One via wallet_id**
fact_wallet_balance → dim_user	**One-to-One via user_id**
fact_wallet_balance → dim_date	**Many-to-One via updated_date_id**
fact_wallet_balance.last_transaction_id	**Degenerate dimension.** Carries the identifier of the most recent transaction that updated the wallet balance. Used for

	auditing, lineage, reconciliation and cross-referencing with **fact_transaction**. No referential join to **fact_transaction**
fact_wallet_balance.last_event_id	**Degenerate dimension.** Carries the identifier of the business event that most recently updated the wallet balance. Used for auditing, lineage and cross-referencing with **fact_user_event**. No referential join to **fact_user_event**.

8. **Source Data Quality Validation Rules**

The FinFlow data generation framework includes both valid and intentionally injected invalid records to simulate real-world data quality issues.

The following validation rules define the expected business and technical constraints applied during data quality testing. Validation checks are used to identify and quantify records that violate these rules.

8.1. Entity Integrity

Validation checks include:

- Primary keys should be unique.
- Primary keys should not be null.
- Reference datasets should not contain duplicate records.

Intentional test scenarios may include:

- Duplicate primary keys
- Null primary keys
- Duplicate reference records

8.2. Referential Integrity

Validation checks include:

- All foreign keys should reference valid parent records.
- Transaction, event, and investment records should have valid parent entities.

Intentional test scenarios may include:

- Orphaned transaction records
- Orphaned investment records
- Invalid dimension references

8.3. Business Rule Validation

Validation checks include:

- Users should be at least 18 years old.
- Signup date should occur after date of birth.
- Wallet activation date should occur after wallet creation date.

- Investment maturity date should occur after investment start date.
- Transaction amounts should be greater than zero.

Intentional test scenarios may include:

- Underage users
- Invalid date sequences
- Negative transaction amounts
- Missing activation dates

8.4. Event Sequence Validation

Validation checks include the following expected progression:

8.4.1. Onboarding Event Sequence Validation

- signup_completed
- → app_login
- → kyc_completed

8.4.2. Account Activation Event Sequence Validation

- → app_login
- → wallet_funded: success
- → wallet_funding_failed: failure

8.4.3. Continuous User Engagement Event Sequence Validation

- → app_login
- → review_plan_options / review_current_investment

8.4.4. New Investment Creation Event Sequence Validation

- → app_login
- → review_plan_options
- → plan_selected
- → savings_plan_created / investment_plan_created

8.4.5. Automated Investment Maturity Event Sequence Validation

- → investment_vests/assets_sale
- → investment_proceeds_wallet_transfer

8.4.6. User Triggered Investment Maturity Event Sequence Validation

- → app_login
- → review_current_investment
- → request_early_withdrawal / assets_sale
- → investment_proceeds_wallet_transfer

8.4.7. Wallet Withdrawal Event Sequence

- → app_login
- → wallet_withdrawal: success

- → withdrawal_failed: failure

Intentional test scenarios may include:
- Wallet funding before KYC completion
- Plan creation before wallet funding
- Missing prerequisite events

8.5. Data Freshness

Validation checks include:
- Daily incremental generation should be completed successfully.
- Event timestamps should not occur in the future.
- Transaction timestamps should not occur in the future.

Intentional test scenarios include:
- Future-dated events
- Future-dated transactions
- Missing daily partitions

8.6. Distribution Validation

Validation checks include monitoring expected distributions for:
- Customer persona assignment
- Acquisition channels
- Product adoption rates
- Funnel conversion rates

Intentional test scenarios include:
- Persona distribution drift
- Acquisition channel skew
- Abnormally high or low conversion rates